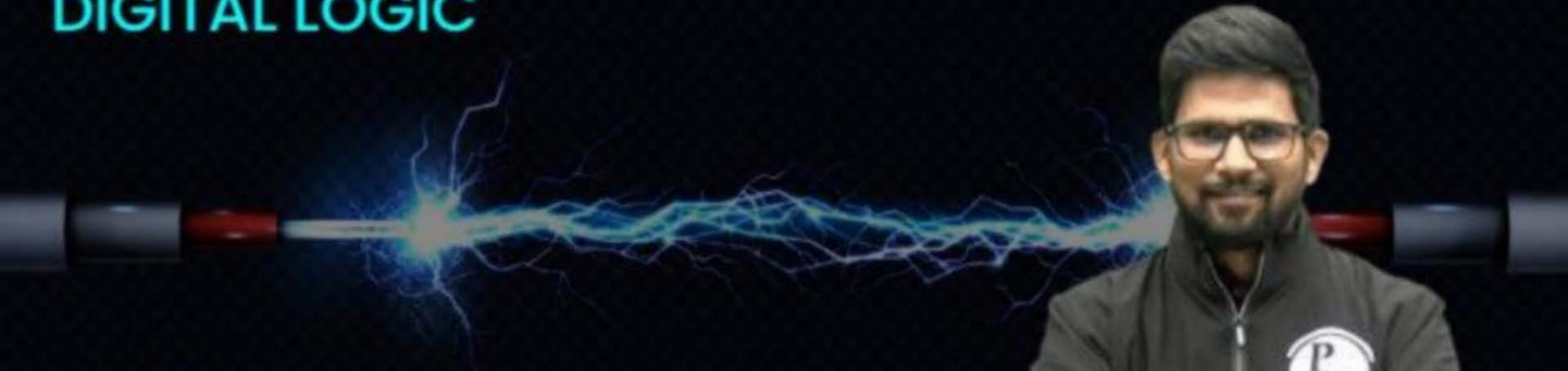
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 09

Combinational Circuit



By- Chandan Gupta Sir



Comparator CKt.



MUX.

Question

We have two number A and B both A and B are 2-bit numbers then in how many combination A > B.

(a) 5

(b) 6

(c) 7

(d) 8

H.W.

Ans. (B).

Question

We have two 4-bit number A and B then number of combinations in which A < B 120

Number of combinations in which A = B 16.

H.W.

H.W.

Q.1. A = 6 bits

B = 2 bits

$N_1 (A > B)$ ____246____

$N_3 (A = B)$ ____4____

Q.2. A = 8 bits

B = 12 bits

No. of min-term (SOP-term) in logical O/P of $W (A = B)$ ____2^8____.

$Z (A < B)$ ____$2^{20} - 2^8 - (255 \times 128)$____

Question

A logic circuit implements the Boolean function :

$$f(A, B, C) = A\bar{B} + \bar{A}\bar{B}\bar{C}$$

$(0,4,5) \quad = \Sigma(0,4,5) + d\Sigma(2,6)$

$100 \quad 000$

It is found that the input combination B = 1 and C = 0 can never occur. Then the simplified output $f(A, B, C)$ will be

(a) ~~$\bar{B}\bar{C} + \bar{A}\bar{C}$~~ $\quad \bar{C} + A\bar{B}$

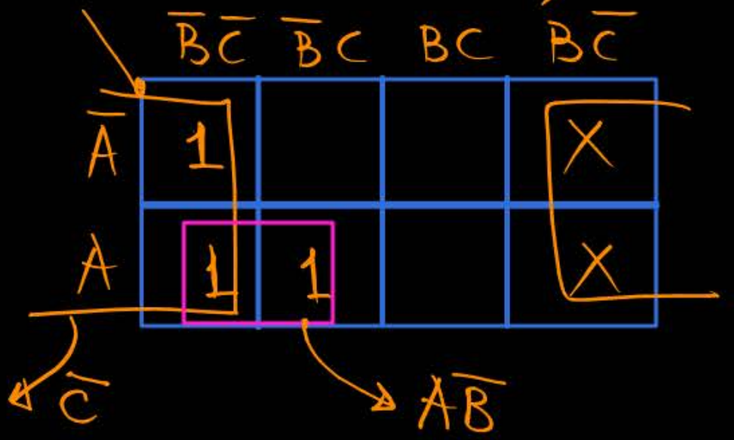
(b) $(A + \bar{C})(\bar{B} + \bar{C}) = \bar{C} + A \cdot \bar{B}$

(c) ~~$\bar{A}\bar{B} + \bar{B}\bar{C}$~~

(d) None of these

	B	C	
0	1	0	→ 2
1	1	0	→ 6

} → don't care

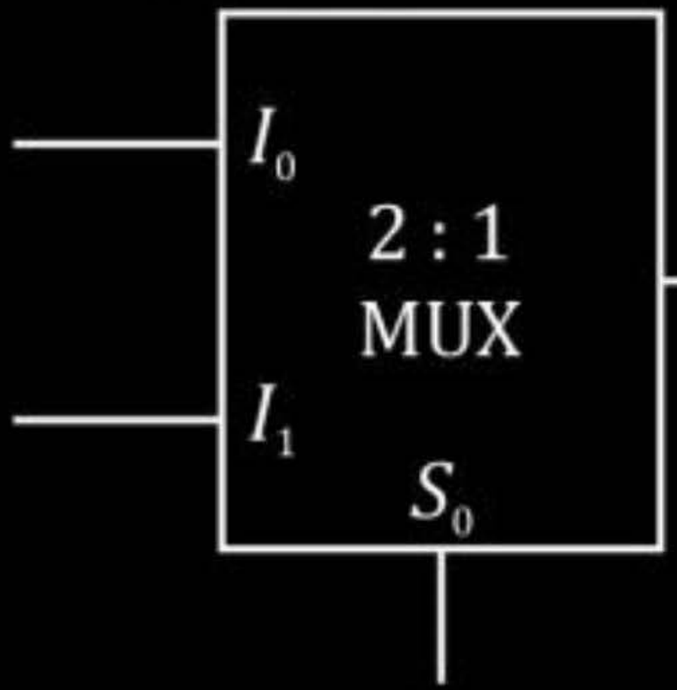


MUX

- What is MUX ?

- It is combinational ckt having many i/p-lines and one-o/p line. And on the basis of select-i/p line, one of the i/p line is transferred to the o/p line.

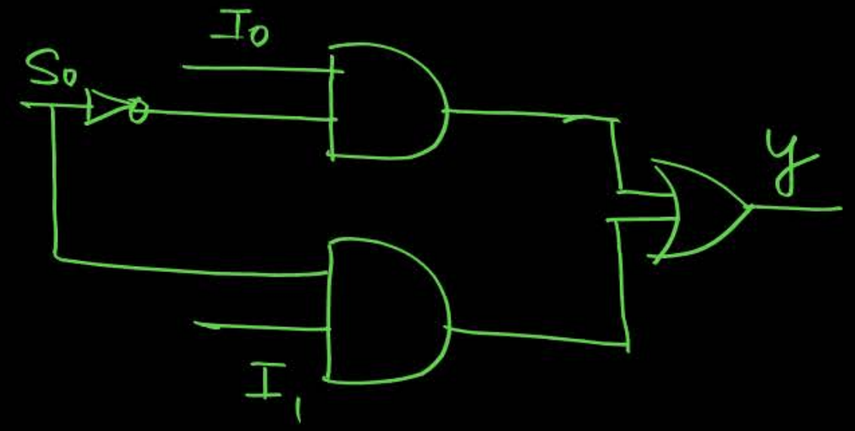
- 2 : 1 MUX



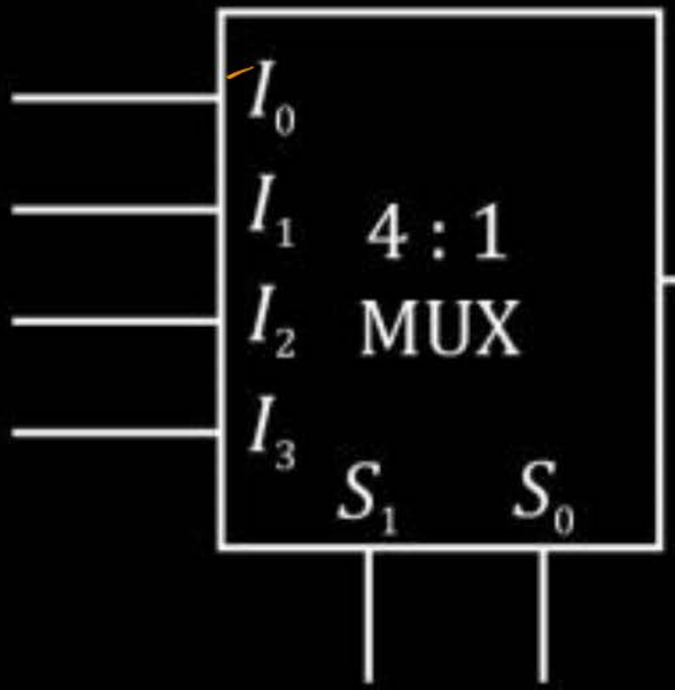
$$y = \overline{S_0} \cdot I_0 + S_0 \cdot I_1$$

$S_0 = 0 \Rightarrow y = I_0$

$S_0 = 1 \Rightarrow y = I_1$

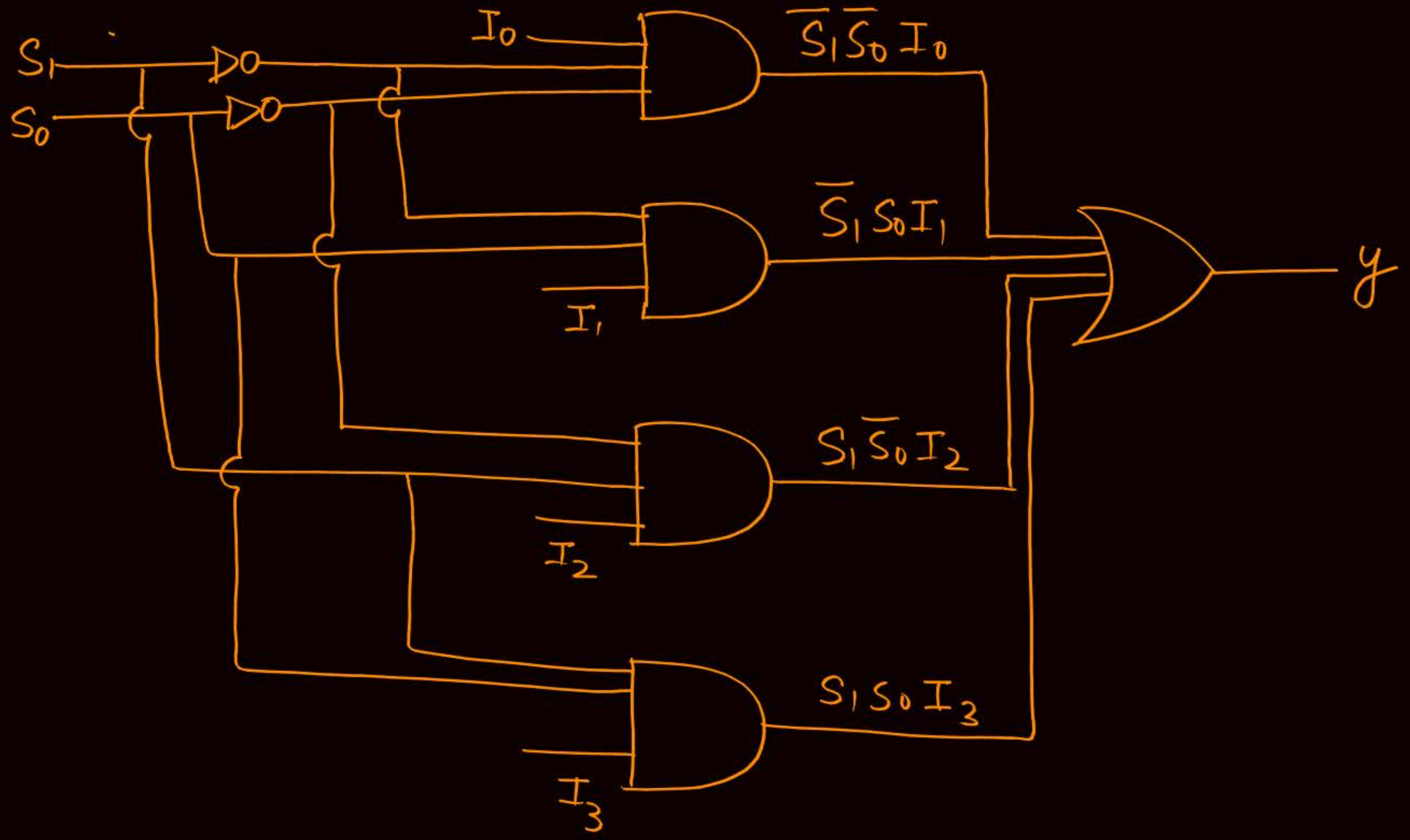


- 4 : 1 MUX

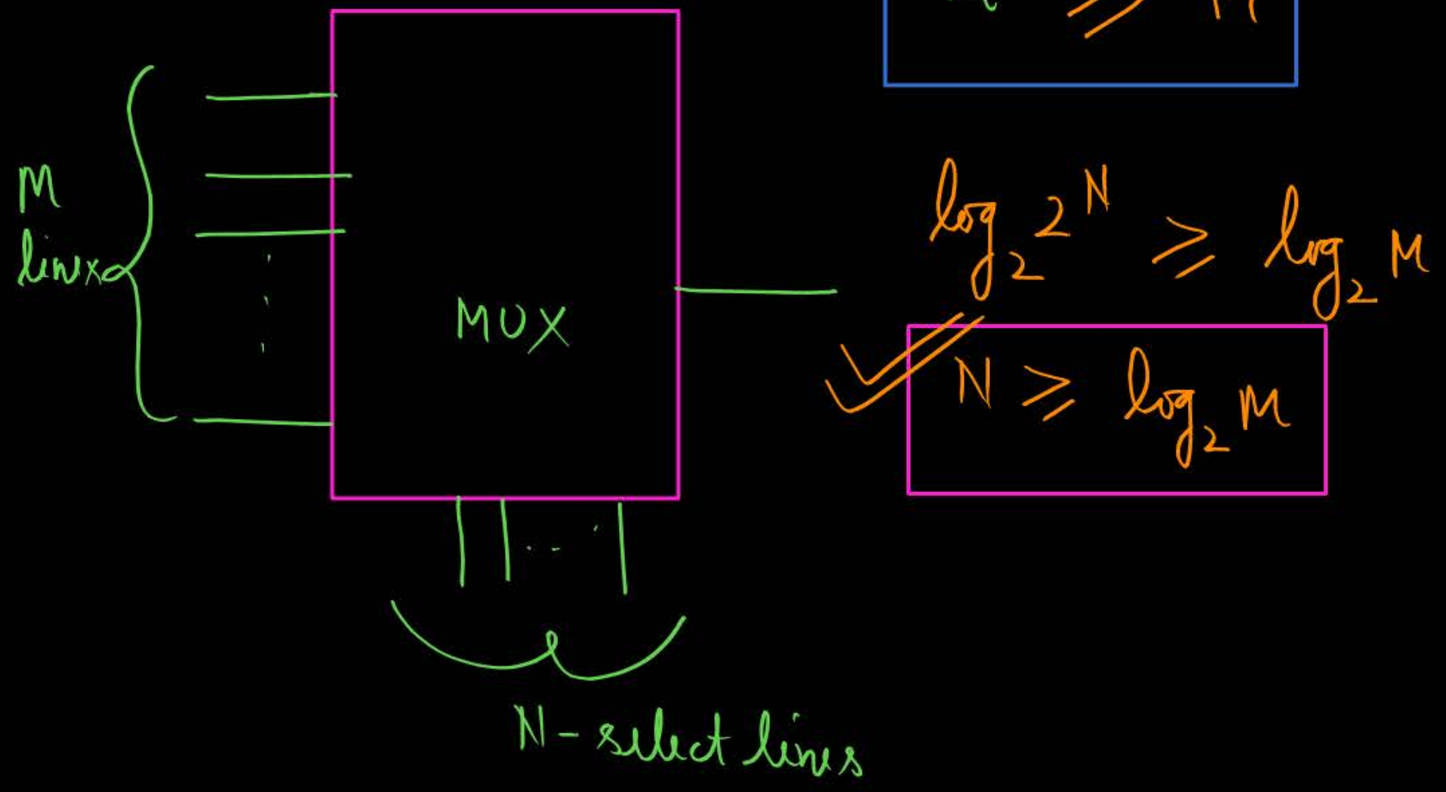


$$y = \overline{S_1} \cdot \overline{S_0} \cdot I_0 + \overline{S_1} S_0 \cdot I_1 + S_1 \overline{S_0} \cdot I_2 + S_1 S_0 \cdot I_3$$

$S_1 = 1, S_0 = 0 \Rightarrow y = I_2$


S_1
S_0
I_0
$\overline{S}_1\overline{S}_0 I_0$
$\overline{S}_1 S_0 I_1$
I_1
$S_1\overline{S}_0 I_2$
I_2
$S_1 S_0 I_3$
I_3
y

- Relation between number of input lines M and number of select lines N :



Higher order MUX using lower order MUX

- 4 : 1 $\xrightarrow{\text{Using 2 : 1 MUX}}$ $2 + 1 = (2^2 - 1)$

  $2^2 : 1$

- 8 : 1 MUX $\xrightarrow{\text{Using 2 : 1 MUX}}$ $4 + 2 + 1 = (2^3 - 1)$

  $2^3 : 1$

- 16 : 1 MUX $\xrightarrow{\text{Using 2 : 1 MUX}}$ $8 + 4 + 2 + 1 = (2^4 - 1)$

  $2^4 : 1$

- 2^n : 1 MUX $\xrightarrow{\text{Using 2 : 1 MUX}}$ $(2^n - 1)$

- 16 : 1 MUX $\xrightarrow{\text{Using 4 : 1 MUX}}$ 4 + 1
- 64 : 1 MUX $\xrightarrow{\text{Using 4 : 1 MUX}}$ 16 + 4 + 1
- 64 : 1 MUX $\xrightarrow{\text{Using 8 : 1 MUX}}$ 8 + 1
- 256 : 1 MUX $\xrightarrow{\text{Using 16 : 1 MUX}}$ 16 + 1

32 : 1 MUX $\xrightarrow{4:1\ MUX}$ (8 + 2)

10 (4:1) MUX + 1 (2:1) MUX

or 11 (4:1) MUX but one 4:1 MUX is used as 2:1 MUX

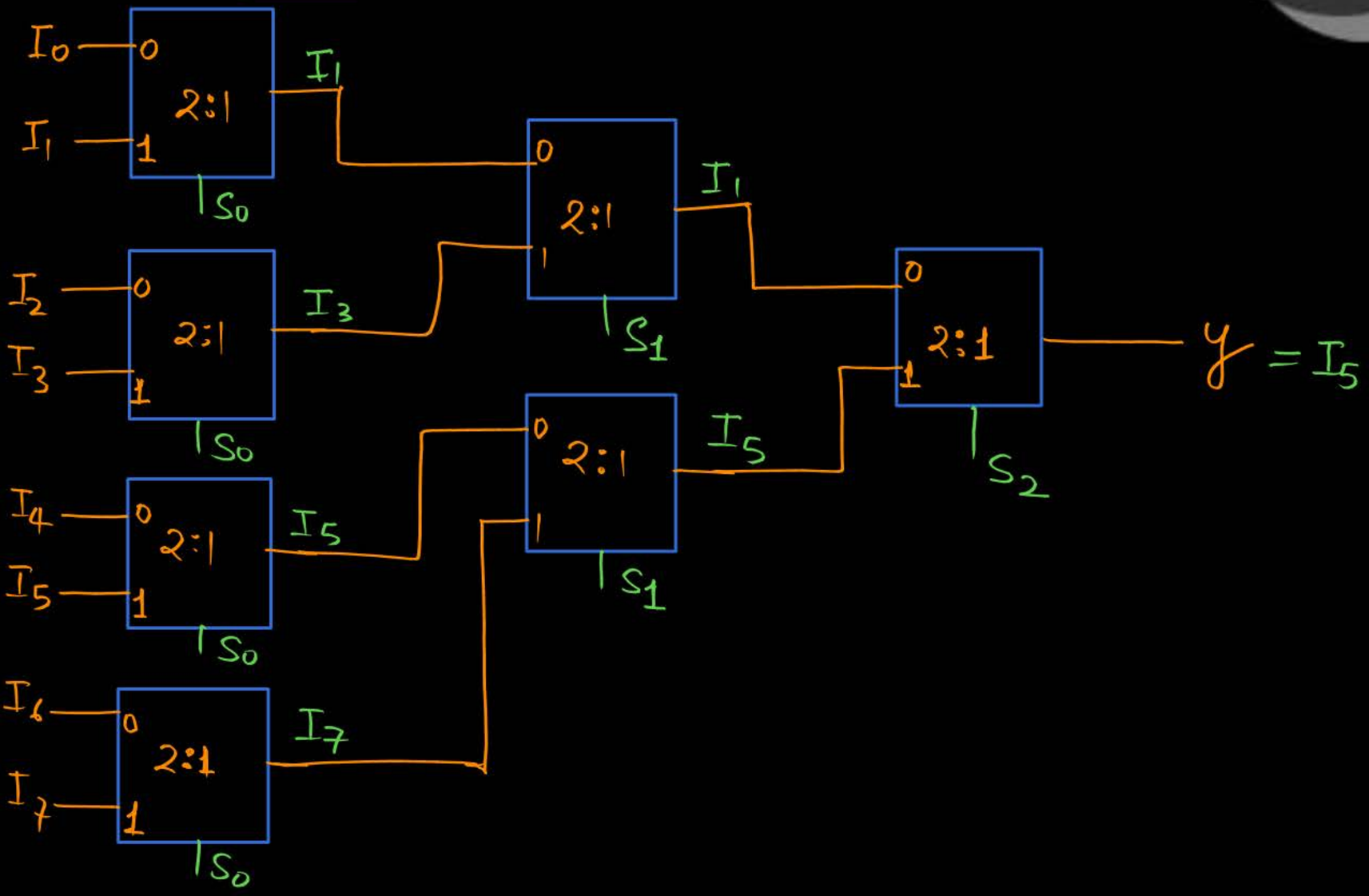
Implement 8 : 1 MUX using 2 : 1 MUX and design the complete circuit.

$I_0 - I_7$ → available

S_2, S_1, S_0

$8:1 \xrightarrow{2:1} 4+2+1$

$S_2 S_1 S_0 = 101$

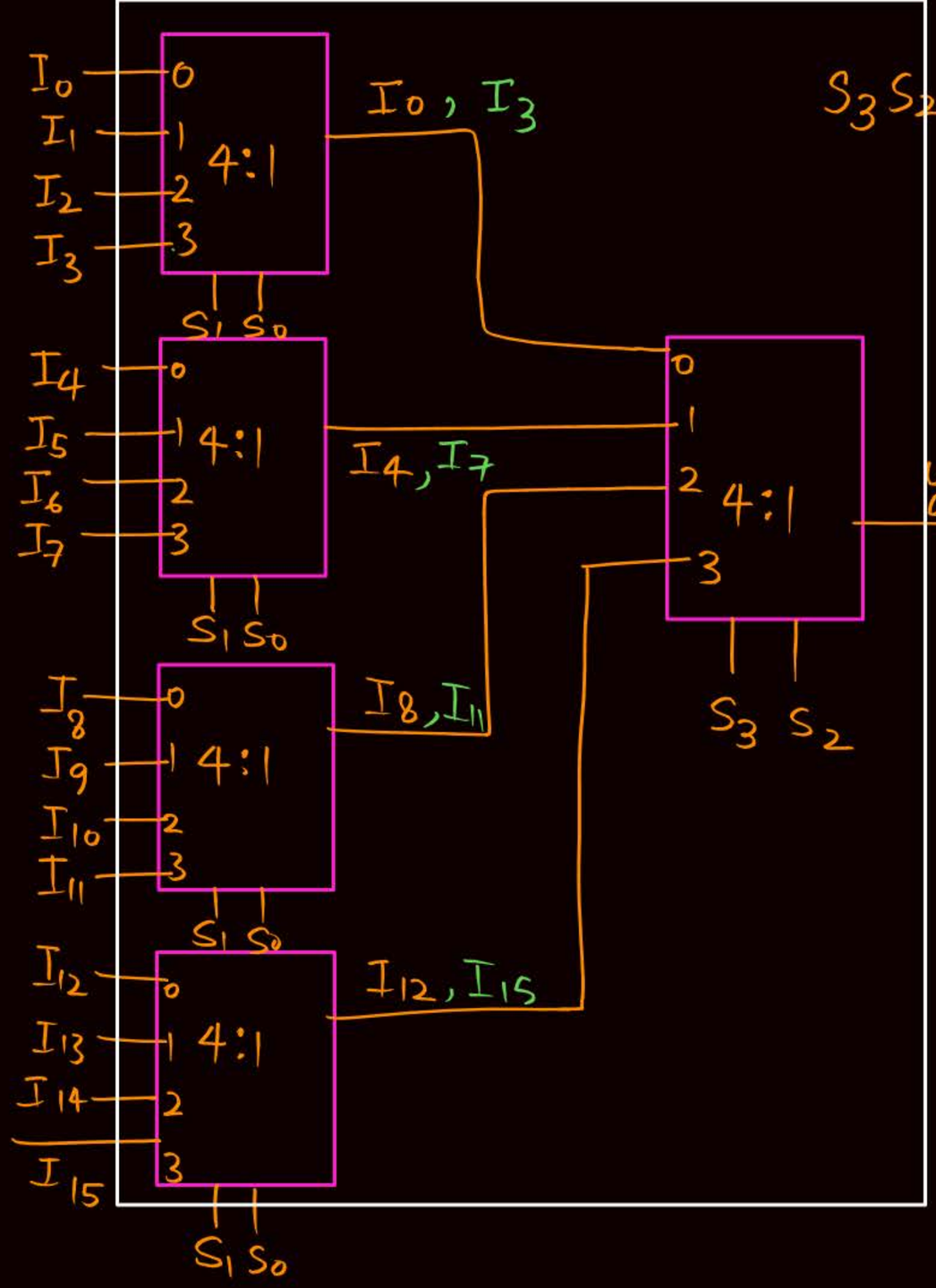


- Implement 16 : 1 MUX using 4 : 1 MUX and design the complete circuit.

$16:1 \xrightarrow{4:1} 4 + 1$

$(I_0 - I_{15})$

$S_3 S_2 S_1 S_0$



$S_3 S_2 S_1 S_0 = (1100)_2 = 12$

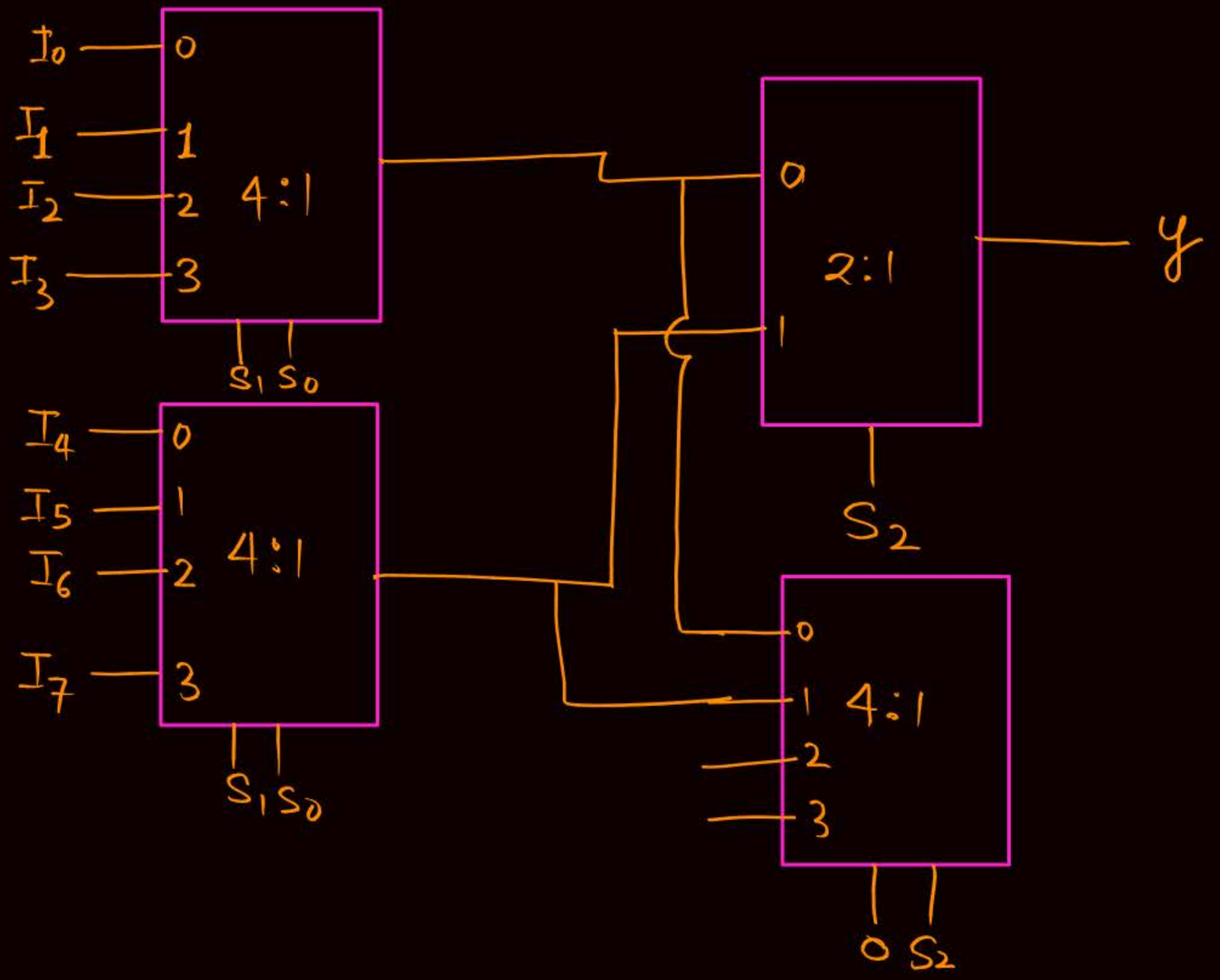
$S_3 S_2 S_1 S_0 = (0111)_2 = 7$

8:1 $\xrightarrow{4:1}$ 2(4:1)MUX + 1(2:1) MUX

or 3(4:1) MUX but one 4:1 MUX will be used as a(2:1)MUX

$(I_0 - I_7)$

S_2, S_1, S_0



- 16 : 1 MUX $\xrightarrow{\text{Using 8 : 1 MUX}}$ 2 (8:1) MUX + 1 (2:1)

- 32 : 1 MUX $\xrightarrow{\text{Using 4 : 1 MUX}}$ (8 + 2) 10 (4:1) MUX + 1 (2:1) MUX.

256:1 MUX $\xrightarrow{8:1}$ (32 + 4) 36 (8:1) + 1 (4:1) or 37 (8:1) MUX

or 11 (4:1) MUX

but one 8:1 MUX will be used as 4:1

H.W.

Q. Design 16:1 MUX using 8:1 MUX.

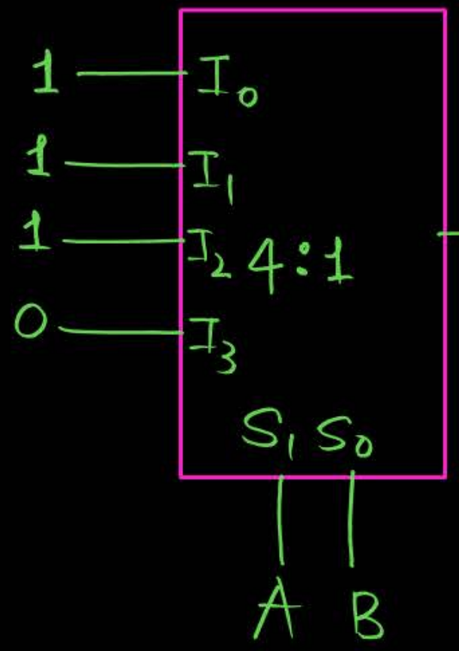
Implementation of Boolean function using MUX

Q. $f(A, B) = \Sigma(0, 1, 2)$ → implement this logical function

(i) using 4 : 1 MUX

(ii) using 2 : 1 MUX

(i).



$f(A,B) = \bar{A}\bar{B}\cdot 1 + \bar{A}B\cdot 1 + A\bar{B}\cdot 1 + AB\cdot 0$

$= \bar{A}\bar{B} + \bar{A}B + A\bar{B}$

$00 \quad 01 \quad 10$



2 Minute Summary

MUX

Thank you

GW

Soldiers !

